Identifying Supply and Demand Shocks Using Dynamic Principal Component Analysis

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Abstract

The article shows that the economy can be described by two factors: an inflationary demand factor and a deflationary supply factor. Our approach combines several filtering techniques to extract business cycle fluctuations and dynamic principal components analysis (DPCA) to extract common factors. The study is performed on the US data (FRED-QD dataset) ranging from 1959 (Q4) to 2022 (Q3). We illustrate that the demand factor is the most important, but the relative importance of each of the two factors changes over time. In addition, we show that the factors obtained using DPCA generally correspond to the alternative identification of supply and demand shock using revisions to expectations from the Survey of Professional Forecasters (SPF).

C10, E32

Dynamic Principal Component Analysis, Supply and Demand Shocks.

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1 Introduction

Dimensionality reduction methods are often used in empirical economics to describe a large amount of data. Most of the studies based on these methods show that the entire structure of the economy can be characterized by using a very small number of factors. In other words, a small number of shocks are the driving forces in the economy. Our article is a continuation of these studies, while we place the main emphasis on the interpretation of these factors. For this reason, we also focus only on the business cycle fluctuations.

One of the main questions in this line of research is how many common factors/shocks are needed to explain the movement of variables in the macro-economy and what are these common shocks, i.e., an 'structural' interpretation of these factors. Examples of studies that ask this or similar questions are, for example, studies by Sargent & Sims (1977), Geweke & Singleton (1981), Forni & Reichlin (1998), Forni *et al.* (2000), Bai & Ng (2002), Giannone *et al.* (2004), Hallin & Liška (2007), Bai & Ng (2007), Forni *et al.* (2009), Hamilton & Xi (2022), Avarucci *et al.* (2022) and many others. Regarding the number of common shocks, the vast majority of these studies agree that the entire economy can be described using a very small number of factors.[1] A widespread opinion is that only two factors describe the whole economy, where one-factor loads upon real variables and the second loads upon nominal variables. For example, this opinion is held by Sargent & Sims (1977), Giannone *et al.* (2004). In contrast, Andrle *et al.* (2017) presents a view that there is only one common factor regarding business cycle fluctuations. The other factors generally represent long-run trends mainly related to inflation and interest rate (expectations) or high-frequency noise. Avarucci *et al.* (2022), on the other hand, presents findings that two shocks are the main driving factors in the economy. The first is an inflationary demand shock, and the second is a deflationary supply shock. These results hold both for business cycle frequencies and also low-frequency movements.

Our study is significantly inspired by the studies mentioned above and shows that regarding business cycle fluctuations, the entire economy can be described by two factors. We dubbed these factors supply and demand given their comovement with GDP and inflation. In more detail, our main contribution is accessing the business cycle fluctuations of a large dataset of US macroeconomic data (FRED-QD dataset) and identifying the main factors affecting selected variables' movements in these frequencies. Practically, we pre-filter data using various filtering techniques to extract business cycle fluctuations. We then apply dynamic (functional) principal components analysis (DPCA) to extract common factors in these frequencies. The use of this method is motivated by the need to capture the lead-lag relations in the data.

Regarding the interpretation of extracted common factors, our identification is generally straightforward: If the factor affects the GDP and inflation in the same direction, we assume that this factor represents a demand shock. On the other hand, if the factor moves the GDP

[1] We abstract here from the division between static and dynamic factors. The term static factor model refers to the static relationship between the data and the obtained factor at time t, but this static factor can be a dynamic process. Therefore, the number of dynamic factors (true shocks) can be smaller than the obtained number of static factors. See, for example, Bai & Ng (2007) for more details.

and inflation in opposite directions, we take that the supply factor affects the economy. Our paper is in this regard also similar to the study by Bekaert *et al.* (2021), which identifies demand and supply shocks based on higher-order moments in the data.

Also, the proposed interpretation of extracted common factors is further supported by the fact that obtained factors correspond to the alternative identification of supply and demand shocks using revisions to expectations from the Survey of Professional Forecasters (SPF) regarding inflation and GDP/GNP forecasts, as will be explained later in detail. In general, it is important to recognize how similar the results of the two fundamentally different methods are. Their high similarity suggests that derived DPCA factors are a good proxy for 'real' supply and demand shocks. (In reality, the supply and demand shocks consist of 'true' structural shocks). This is due to the fact that this alternative identification method uses a novel methodology that is most often used to identify monetary policy shocks. When forecasters do not make systematic errors in their predictions, revisions to expectations can be considered 'true' shocks. On the other hand, as will be shown later, revisions to expectations can generally be quite noisy. This is not the case regarding DPCA factors, as these factors are obtained using a large amount of data. However, the extraction method used makes the interpretation of these factors much less clear. For that reason, if both methods give similar results, this supports the supply and demand interpretation of the DPCA factors, which goes beyond simple regression/correlation analysis. In other words, our results suggest that obtained factors correspond to supply and demand shocks perceived by professional forecasters. At the same time, it is also an important finding on its own how practitioners identify supply and demand shocks regarding business cycle fluctuations, i.e., how they revise their forecast in time.

Our results generally support findings by Avarucci *et al.* (2022) and, to some extent, findings by Andrle *et al.* (2017). The economy, in our view, can be described by two factors: inflationary demand factor and deflationary supply factor. As will be explained later in detail, the inflationary demand shock is much more prominent and explains most of the variation in business cycle fluctuations. However, the contribution of the deflationary supply shock is non-negligible, especially regarding inflation data. These are important findings, given the fact that the practical thinking of many practitioners is still dominated by the IS-LM and AD-AS framework and the associated thinking about the business cycle in terms of supply and demand shocks. The obtained empirical results also have significant implications for theoretical literature. Theoretical models should account for the fact that a small number of factors exist in the economy and that the distribution of these factors is non-gaussian. We stress that both facts should be accounted for in the theoretical models. For example, this result can be achieved by models with sufficiently prominent amplification mechanisms.

2 Data and Methodology

The article uses a large dataset obtained from McCracken & Ng (2020), which generally describes the structure of the US economy as it includes a majority of macroeconomic and financial series. In more detail, we use a 248 quarterly time series dataset from 1959 (Q1) to 2022 (Q3).

This dataset generally replicates the Stock & Watson (2012) original dataset and provides the broadest possible number of indicators on a quarterly frequency. The series provided can be divided into 14 groups: NIPA; Industrial Production; Employment and Unemployment; Housing; Inventories, Orders, and Sales, Prices; Earnings and Productivity; Interest Rates, Money and Credit, Household Balance Sheets, Exchange Rates, Other, Stock Markets, and Non-Household Balance Sheets. A complete list of the data is given in the appendix in McCracken & Ng (2020).

The data are provided in levels, i.e., in their original form without any transformation. As a consequence, many of the series are non-stationary. However, the authors provide benchmark transformation codes to stationarize the time series.[2] However, we do not follow this benchmark; instead, we rely on a similar approach presented by Hamilton & Xi (2022) or Avarucci *et al.* (2022). Regarding the series that are transformed according to McCracken & Ng (2020) using logs, first differences of logs, or second differences of logs (transformations 4-6), we take the log transformation of the given variables. Then we directly apply one of the below-mentioned filters to extract cyclical components. Regarding the series used in levels, first differences, or second differences (transformations 1-3), we use the variables in level and then extract cyclical components. This is because we want to avoid over-differencing.[3] Another change relative to McCracken & Ng (2020) is that we take interest rates in levels rather than logs and take the first difference regarding price indices before filtering the series. This is because, even before the formal adoption of inflation targeting, central banks focused on inflation (gap) rather than the price level. Therefore, we are interested in cyclical fluctuations around the inflation 'target'.

To extract business cycle fluctuations directly, we use the Christiano-Fitzgerald filter (Christiano & Fitzgerald, 2003), which is a band-pass filter. Regarding the Christiano-Fitzgerald filter, we focus on frequencies between 6-32 quarters. As a robustness check, we also use the Hodrick-Prescott filter (Hodrick & Prescott, 1997) with lambda equal to 1600, which is a high-pass filter, and Hamilton filter (Hamilton, 2018), which amplifies specific frequencies while completely canceling others (Schüler, 2018). Although this approach may appear less robust than directly applying DPCA, this approach allows simple extraction and identification of factors in the time domain, which significantly simplifies interpretation. At the same time, working with the time domain allows us to visualize the changing importance of individual factors in time, as will be shown later. It is also important to stress that we standardize the data to have a mean of zero and a unit standard deviation.

One caveat of this dataset is that not all the series are available from the beginning, or values can be missing, i.e., the dataset is unbalanced. Due to this fact, we use the approach offered by McCracken & Ng (2020)[4] to impute missing values. This method is an expectation maximization approach algorithm, i.e., we iteratively extract principal components to impute missing values.

[2]The benchmark transformation codes indicate whether the series should be transformed by taking logs, taking first or second-order differences, or combining the above-mentioned transformations. Due to this transformation, all series can be considered stationary.

[3]It should be noted that the difference transformation is a filter that highlights certain frequencies in the data.

[4]This approach is itself adapted from Stock & Watson (2002).

Our preferred method for extracting common factors is dynamic principal components analysis. DPCA, as mentioned in the introduction, is a dimensionality reduction method, and it is based on the eigenvalue decomposition of the spectral density matrix. DPCA allow us to decompose observed time series $x_{i,t}$ in following way:

$$x_{i,t} = \sum_{k=1}^{q} b_{k,t}(L) u_{k,t} + \xi_{i,t} \tag{1}$$

where $u_{k,t}$ denote the common factor, $b_{k,t}(L)$ denotes one-sided linear filters, q denotes the number of common factor[5], and $\xi_{i,t}$ denotes idiosyncratic noise, which is uncorrelated with the common factor. Also, we only allow 'weak' correlation among $\xi_{i,t}$. In the time-domain, DPCA can be represented as a two-sided filter, which allows us to filter the common factor. Therefore, in the time-domain DPCA can be written in the following way:

$$u_{k,t} = \sum_{k=-L}^{L} \sum_{i=1}^{n} w_{i,k} x_{i,t+k} \tag{2}$$

where $w_{i,k}$ denotes the filter weights regarding individual series, L denotes the number of lags and leads used. We use two lags/leads ($L = 2$) to account for lead-lag relationships, but the results are generally robust. For more formal treatment of DPCA, see Forni *et al.* (2000), or Hörmann *et al.* (2015).

For a robustness check, we also use data from the Survey of Professional Forecasters (SPF) currently conducted by the Federal Reserve Bank of Philadelphia. Regarding SPF data, we use median value (growth rate) data for the following series: GDP/GNP and GDP/GNP price deflator.[6] We use all available data, i.e., the period from 1969 (Q4) to 2022 (Q3), i.e., the dataset is somewhat shorter than the dataset obtained from McCracken & Ng (2020). The data are again available only on a quarterly frequency.

SPF data provides us with an alternative way to identify supply and demand shocks, and we are specifically interested in how similar these shocks are to the factors obtained using PDCA.[7] We identify these shocks using the revision to expectations in the inflation and GDP/GNP. In more detail, if there is an upward revision of expectations about GDP/GNP and inflation, we assume that a (positive) demand shock prevails in the economy in the given period. On the other hand, if there is a revision of the GDP/GNP and inflation, which is in the opposite direction, we assume that supply shocks prevail. Practically, to determine revisions to expectations, we use the difference between the expected value for the quarter in which the Survey is conducted. We compare this value with the expectations for this quarter in the previous round of the Survey, i.e., $E_t x_t - E_{t-1} x_t$, where x_t is a (forecast of) either GDP/GNP (y_t) or inflation (π_t). The Survey sets the deadlines for responses in the second to the third week of the middle month

[5] A set of n time series can always be explained by n common factors. However, we assume that $q < n$.

[6] Note that research by Mboup & Wurtzel (2018) shows that there are basically no differences between the mean and median forecasts for real GDP and inflation.

[7] Given that we rely only on the differences of the expected values, these are rather to be considered perceived shocks as they do not necessarily have to coincide with the real shock that in the economy.

of each quarter when the Survey is conducted. Therefore, the forecasters' information datasets do not necessarily coincide with the historical data. Due to this fact, we use data for the most recent forecast periods, which should be the least sensitive to changes in expectations.

In terms of methodology, we generally follow Jarociński & Karadi (2020) and especially Jarociński (2022)[8], where authors distinguish monetary shocks according to the high-frequency response of the stock market and fed fund futures. Instead, we use the revisions to the expectations regarding output and inflation, assuming that different types of shocks affecting the real economy cause different revisions of expectations. We specifically compute the "rotational sign restrictions" approach.[9] This allows for supply and demand shock to hit the economy simultaneously. The decomposition satisfies the following:

$$M = UC \tag{3}$$

where $M = (E_t y_t - E_{t-1} y_t, E_t \pi_t - E_{t-1} \pi_t)$ is a $T \times 2$ matrix with shock regarding revisions regarding GDP $(E_t y_t - E_{t-1} y_t)$ in the first column and revisions to inflation $(E_t \pi_t - E_{t-1} \pi_t)$ in the second column. T denotes the length of the dataset. $U = (shock_{supply,1:T}, shock_{demand,1:T})$ is again a $T \times 2$ matrix with $shock_{supply,1:T}$ in the first column and $shock_{demand,1:T}$ in the second column and the columns are mutually orthogonal. Therefore U satisfy following condition: $U^T U = I$. C is a 2×2 matrix, which captures how revisions to inflation and GDP translate into supply and demand shock. The first column of C contains a vector of ones, which reflect the fact that $shock_{total,t} = shock_{demand,t} + shock_{supply,t}$, i.e., the economy is simultaneously hit by supply and demand shocks. The second column of C contains the elasticities of inflation revisions to $shock_{supply,1:T}$ and $shock_{demand,1:T}$, where $shock_{supply,1:T} < 0$ (first row) and $shock_{demand,1:T} > 0$ in the second row. Also, note that this decomposition is not unique, as multiple combinations of U and C satisfy the sign restrictions. We specifically use the median target rotation, which means using the median of the admissible rotation angles. This suggest that shares of variance of $shock_{total,1:T}$ explained by $shock_{demand,1:T}$ and $shock_{supply,1:T}$ is the same.

3 Empirical Results

In this section, we present the results. We first estimate the dynamic principal components and present their summary statistics. Subsequently, based on these results, we described the co-movement of selected variables with obtained factors. Subsequently, we decompose GDP and inflation movements in the time domain. Then, we use these obtained shocks to study the impulse response function of selected variables regarding these shocks. For a robustness check, we also study how similar the shocks obtained from SPF data are to the factors obtained using DPCA.

[8]For a full description of the methodology used, see the original paper by Jarociński (2022). We only slightly modified the original approach.

[9]Since we are interested in how similar the shocks obtained by this method are to the factors obtained using PDCA. We cannot directly divide supply and demand shock based on the sign restrictions, i.e., $shock_{supply(demand),t} = (E_t \pi_t - E_{t-1} \pi_t)$ if $(E_t y_t - E_{t-1} y_t) \times (E_t \pi_t - E_{t-1} \pi_t) < (>) 0$, else 0. Because this approach only allows one type of shock to be present in the economy in a given period.

The summary statistics of the first three dynamic principal components obtained are shown in Table 1. The individual columns in the table display (in the following order) the variance explained, cumulative variance explained, standard deviation and skewness, and kurtosis of each factor. The following two columns display p-values of the statistical tests of non-normality. More specifically, we use Jarque-Bera and Kolmogorov–Smirnov tests to test for non-normality. It can be seen that the explanatory power of the first factor is 38 %, which clearly dominates over the other factors. The distribution of this factor is left-skewed, and it can be considered leptokurtic. (This is an empirical distribution of the factor collected from time 0 to time T, i.e., whole sample distribution.) As will be discussed below, we identify the first factor as a proxy for a demand shock. Given this fact, it makes intuitive sense that there is a much more limitation regarding upward demand shock, i.e., production function constraint. However, the severe negative demand shock is much more plausible. Also second and third factor seems to have some explanatory powers compared to the rest of the factors. The second factor explains 15 % of the variance, and the third factor explains 10 % of the variance. Explanatory powers of other factors are practically negligible. The distribution of the second factor is very slightly left-skewed, and it can be again considered leptokurtic. Below, we explain why we consider the second factor a proxy for supply shocks. However, it again makes intuitive sense that supply shocks, i.e., shocks to productivity, will be much less skewed. Therefore, our results support that the common factors are non-Gaussian.

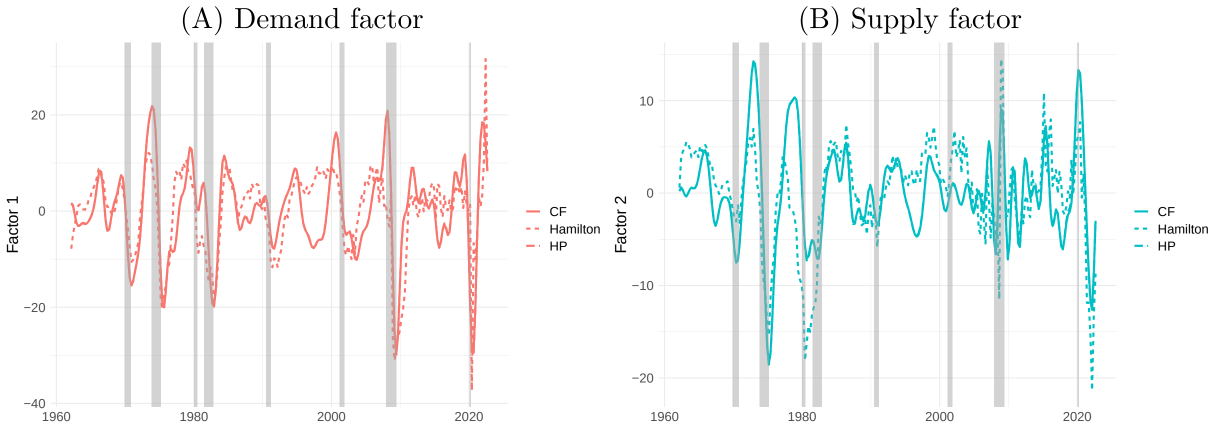
Table 1: Summary Statistics - Factors

	Var	Cum.Var	Std.Dev	Skewness	Kurtosis	Jarque-Bera	Kolmogorov–Smirnov
CF filter							
F 1	0.38	0.38	9.18	-0.43	3.95	0.00	0.30
F 2	0.15	0.53	5.15	-0.07	4.50	0.00	0.22
F 3	0.10	0.62	2.03	-1.46	9.32	0.00	0.00
Hamilton filter							
F 1	0.31	0.31	8.63	-1.21	5.86	0.00	0.00
F 2	0.14	0.46	5.46	-1.14	4.88	0.00	0.00
F 3	0.08	0.54	3.32	1.96	10.72	0.00	0.01
HP filter							
F 1	0.37	0.37	9.13	-0.44	3.98	0.00	0.36
F 2	0.15	0.53	5.13	-0.10	4.55	0.00	0.25
F 3	0.09	0.62	2.27	0.65	3.92	0.00	0.01

Note: The first and second columns show the variance explained and the cumulative variance explained by each factor. The next three columns display the standard deviation, skewness, and kurtosis of every factor. The next two columns display p-values of the statistical tests of non-normality (Jarque-Bera and Kolmogorov–Smirnov test). CF denotes the Christiano-Fitzgerald filter. HP denotes the Hodrick–Prescott filter.

Figure 1 displays the first two factors in the time domain to gain further intuition on the results. Note that factors obtained using the Hodrick–Prescott filter are almost indistinguishable from those obtained using the Christiano-Fitzgerald filter. Hamilton filter somewhat differs, but the general picture regarding factor development in time is the same. Also, notice that regarding the first factor, we can see sharp contractions that coincide with NBER-dated recessions, which is again in line with our interpretation of this factor as a proxy for demand shock.

Figure 1: Factors



Note: The left panel shows the extracted first (demand) common factor. The right panel shows the extracted second (supply) factor. CF denotes the Christiano-Fitzgerald filter. HP denotes the Hodrick–Prescott filter. Shaded regions denote NBER-dated recessions.

We decided to reduce the dimensionality of our data, given the results above (regarding the variance explained by each factor). We follow Avarucci *et al.* (2022) and extract the first two factors.[10] To gain further insight into the results, we examine the co-movement of factors with individual series. We measure co-movement using a correlation coefficient and unadjusted R-squared. Table 2 shows the results, divided into sections for each filtering method. The first and second columns show the correlation coefficient of individual series with the first or second factor. In the last column, we show the unadjusted R-squared[11] explained by both factors.

Our main findings are as follows: The first factor is mainly linked to the real economy. It explains most of the variation in GDP. However, the factor is also related to nominal variables as it explains about a third[12] of inflation. Given the fact that the factor also positively correlates with inflation and the real economy, we consider it a demand shock. The second factor is mostly related (negative correlation) to nominal variables, i.e., inflation fluctuations. At the same time, it is much less related to the real economy, although it still has some non-negligible explanatory power (positive correlation with real economy variables). For this reason, we consider it a supply factor. Also, it can be seen that the factors loadings remain quite stable when we change the pre-filtering method, which suggests the robustness of our results. The results for categories are shown in the Appendix (Table A1). The table has a structure similar to Table 2. The first and second columns show the average correlation coefficient of the given category with the first or second factor. The following two columns display the average R-squared with the first or second factor. The last column shows the average (unadjusted) R-squared obtained by regressing the

[10] The literature does not provide a clear answer to the number of factors. Aside from Avarucci *et al.* (2022), we also employ information criteria by Bai & Ng (2002) and by Alessi *et al.* (2008) to determine an optimal number of static factors. These indicate that there are between 8-25 factors. These are, however, static factors.

[11] Note that to compute correlation coefficients and R-squared, we use the individual series that are detrended using the same filtering method used to obtain the common factors.

[12] Correlation coefficient squared.

individual series on both factors. The results support the above-mentioned interpretation of the factors.

Table 2: Co-Movement Statics - Selected Variables

	COR_1	COR_2	RSQ_total
CF filter			
CPIAUCSL	0.62	-0.57	0.75
GDPC1	0.91	0.36	0.92
INDPRO	0.91	0.31	0.90
PCECTPI	0.58	-0.60	0.74
PCEPILFE	0.47	-0.72	0.79
UNRATE	-0.91	-0.12	0.84
HP filter			
	COR_1	**COR_2**	**RSQ_total**
CPIAUCSL	0.62	-0.56	0.75
GDPC1	0.91	0.36	0.92
INDPRO	0.91	0.31	0.90
PCECTPI	0.58	-0.60	0.74
PCEPILFE	0.47	-0.72	0.79
UNRATE	-0.91	-0.12	0.84
Hamilton filter			
	COR_1	**COR_2**	**RSQ_total**
CPIAUCSL	0.37	-0.71	0.65
GDPC1	0.90	0.30	0.89
INDPRO	0.90	0.20	0.84
PCECTPI	0.34	-0.76	0.71
PCEPILFE	0.27	-0.77	0.67
UNRATE	-0.85	-0.13	0.73

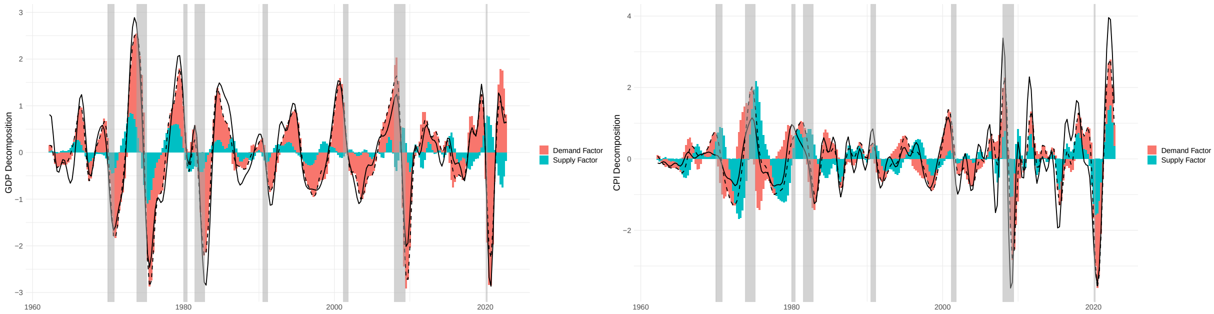
Note: The first and second columns show the correlation coefficient of individual series with the first or second factor. The third column shows the unadjusted R-squared obtained by regressing the individual series on both factors. CF denotes the Christiano-Fitzgerald filter. HP denotes the Hodrick–Prescott filter.

Figure 2 shows (detrended) GDP and CPI decomposition in the time domain using the first two obtained factors. Specifically, we primarily use the first DPCA two factors, GDP (GDPC1) and CPI (CPIAUCSL), which are detrended by the Christiano-Fitzgerald filter. (Both series are also standartized to have mean of 0 and a standard deviation of 1.) Note that this identification corresponds to general intuition. Supply shocks mainly characterize 1970s energy crises. The recession of 1981–82 was essentially a demand (shock) recession as it was caused primarily by monetary policy. The early 2000s recession and 2007–2008 financial crisis can also be characterized as caused mainly by demand shocks. However, the Covid 19 pandemic is described by a series of demand and supply shocks, indicating this period's unclear character. In short, this illustrates that the demand factor is the most prominent driver in the economy, but the relative importance of each of the two factors changes over time.

Results regarding other filtering methods are shown in the Appendix. Figure A1 shows decomposition using variables and factors obtained from the Hamilton filter. Compared to the results obtained from the Christiano-Fitzgerald filter, the second factor seems to explain much more inflation dynamics. Figure A2 shows the decomposition using the Hodrick–Prescott filter detrended variables. This decomposition is almost identical to the Christiano-Fitzgerald filter decomposition as expected.

Figure 2: Decomposition (CF filter)

(A) GDP (B) CPI



Note: The dotted line represents a fitted value of GDPC1/CPIAUCSL. Solid line is detrented GDPC1/CPIAUCSL. Shaded regions denote NBER-dated recessions.

Lastly, we try to identify the time-varying impact of supply and demand shocks. More specifically, we use the obtained factors as instruments to directly estimate their impact on the variables of interest using a local projection method[13] by Jordà (2005). The model is estimated in the following way:

$$x_{t+h} = \alpha + b_h u_{1,t} + c_h u_{2,t} + e_{t+h} \tag{4}$$

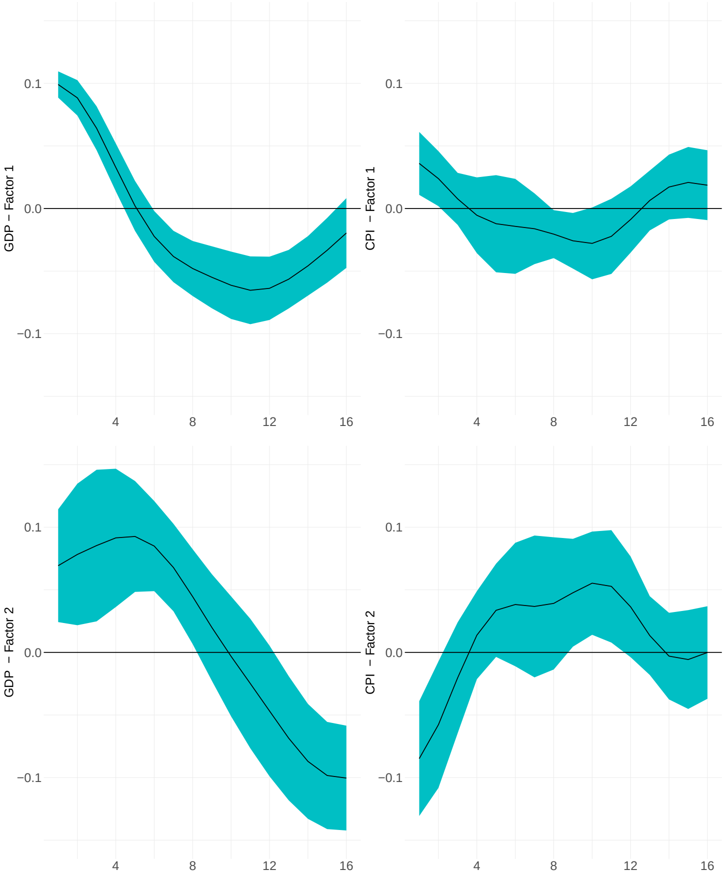
where $u_{k,t}$ again denote common factors[14], x_{t+h} is either GDP gap (GDPC1) or CPI gap (CPIAUCSL). α and e_t denotes constant and error term. b_h and c_h then denote the coefficients of interest, which shows the direct impact of these shocks for periods $h = 0, 2, ...16$, i.e., four years. Also, we would like to stress that the size of impulse responses is again regarding the detrended variables by the Christiano-Fitzgerald filter. As it is standard practice regarding local projection, we use Newey-West robust standard errors. Also, note that we use 95% confidence intervals.

Figure 3 show the impulse response functions of GDP and CPI to the supply and demand shock (first and second factor obtained using Christiano-Fitzgerald detrended data). The top row shows the impulse response functions of GDP and CPI to the first (demand) factor. The bottom row shows the same regarding the second (supply) factor. The results are again in line with general intuition. A demand shock on impact increases inflation and GDP. The effect on GDP gap is much more pronounced. However, this effect vanishes in about a year. After that, it seems that a contraction in the economy occurs, reducing both the GDP and inflation gaps. Regarding the supply factor, we can see that there is an increase in the GDP and a reduction in the CPI upon impact. A positive effect of supply shock on the GDP seems to be longer lasting, approximately for two years. However, the results also suggest that the supply shock starts to have a much less clear impact on inflation after roughly a year. Results regarding other filtering methods are again shown in the Appendix (Figure A3, Figure A4). Both alternative impulse response functions are almost identical to the above-presented results.

[13] This approach is basically event study analysis.

[14] $u_{1,t}$ and $u_{2,t}$ denotes the directly observed factors, which are by definition practically mutually orthogonal.

Figure 3: Impulse response functions (CF filter)



Note: The green bands denote Newey-West robust standard errors (95% confidence intervals). The top row shows the impulse response functions of GDP and CPI to the first (demand) factor. The bottom row shows the impulse response functions of GDP and CPI to the second (supply) factor.

For a robustness check, we use the supply and demand shock obtained from SPF data[15] and compare them to the factors obtained using DPCA. Given the fact that the shocks obtained from SPF data can be considered quite noisy, we also filter them using the Hodrick Prescott filter with lambda set to 1. (We use the trend estimate.) We again primarily use the factors obtained by the Christiano-Fitzgerald filter. The results of other filtering methods are shown in the appendix. Also, note that SPF shocks were re-scaled for visualization purposes to have the same variances as DPCA factors.

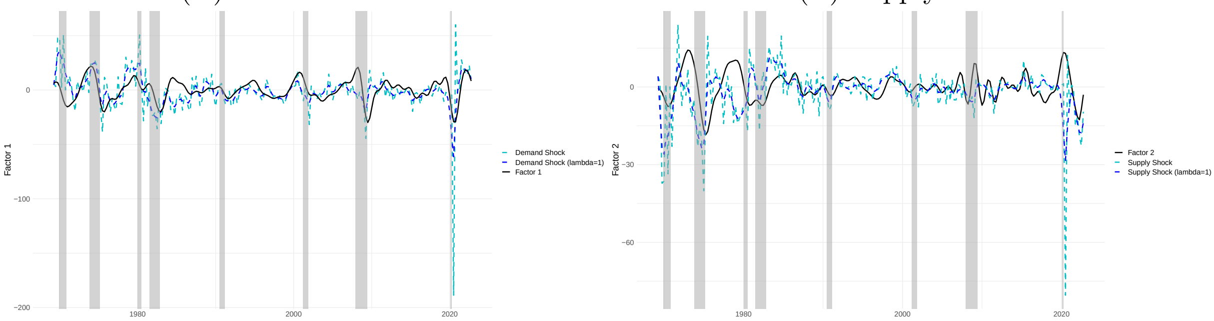
Figure 4 shows the results regarding DPCA factors obtained using Christiano-Fitzgerald detrended data. Results regarding other filtering methods are again shown in the Appendix (Figure A6, Figure A7). The left panel shows extracted demand shocks from SPF data and the first DPCA factor. The right panel shows the supply shocks extracted from SPF data and the second DPCA factor. Our results suggest that the first (demand) factor has similar dynamics to the demand shock obtained from SPF data. This similarity is a bit more pronounced using

[15]Figure A5 in the Appendix display the 'original' revisions to expectations from Survey of Professional Forecasters (SPF) data. Also, note that all results below remain valid if we first regress revisions to expectations on their lagged values, as there may be some rigidities in the forecasts. (Coibion & Gorodnichenko, 2012)

filtered SPF shocks. A bit more formally, the correlation of the first factor with the demand shock obtained from SPF data is 0.32, respectively 0.49 regarding the filtered version of the SPF demand shocks. Regarding the second factor, the similarity between the second factor and supply SPF shock is much less clear. The correlation of the second DPCA factor with the SPF supply shocks is only 0.07, respectively 0.1 regarding the filtered version. However, there appears to be a phase shift in these time series regarding supply shocks. Also, note that SPF data are from each quarter's middle month (second to third week). This suggests that SPF-extracted shocks should be somewhat lagging behind DPCA factors. Regarding other filtering methods, DPCA factors obtained from HP-filtered data are again very similar to the above-described results. On the other hand, it is interesting that the results factors obtained from Hamilton filtered data have practically the highest similarity with the shocks extracted from SPF data. This is especially true of SPF-extracted supply shocks and the second DPCA factor.

Figure 4: Factors (CF filter) - Robustness check

(A) Demand factor (B) Supply factor



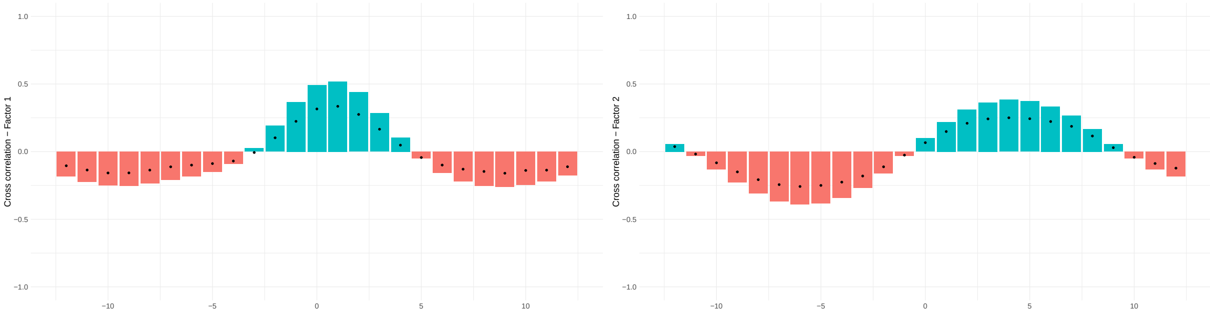
Note: The left panel shows extracted demand shocks and the first DPCA factor. The right panel shows extracted supply shocks and the second DPCA factor. The supply and demand shocks were obtained from SPF data using revisions to expectations. SPF shocks were re-scaled to have the same variances as DPCA factors. Shaded regions denote NBER-dated recessions.

For the reasons regarding possible phase shifts in the series considered, we look more closely at the cross-correlation of these series. Figure 5 shows the results regarding Christiano-Fitzgerald detrended data. Results regarding other filtering methods are again shown in the Appendix (Figure A8, Figure A9). The left panel shows the cross-correlation of the SPF-extracted demand shocks with the first (demand) DPCA factor. The right panel shows the cross-correlation of SPF-extracted supply shocks with the second (supply) DPCA factor. Bars denote the cross-correlation with the HP-filtered series of the supply/demand shocks. Dots denote the cross-correlation with the unfiltered series of supply/demand shocks. Specifically, we estimate the correlation between common factor $u_{k,t+k}$, and $shock_{demand,t}$,$shock_{supply,t}$. The results show that the time series of demand shocks (both filtered and unfiltered) and the first factor are essentially phase aligned (or the first factor is leading by one quarter). On the other hand, there seems to be a phase shift around four quarters (both leads and lags) regarding the time series of supply shocks and the second factor. This points out that professional forecasters are able to identify supply shocks only with a delay. Alternatively, this could be explained by the fact that supply shocks are much less frequent and are, therefore, harder to identify. Or that the propagation mechanism regarding supply shocks is non-trivial, and it is much harder to distinguish between the shocks to trend and the cycle. This is also supported by the fact that the correlation between the CPI gap (CPIAUCSL) obtained by HP/CF filter and the SPF-extracted supply shocks is practically zero. Again, there seems to be a phase shift around four quarters (both leads and lags). As mentioned before, Hamilton-filtered DPCA factors have the highest correlation with the shocks extracted from SPF data. There is also practically no phase shift regarding the SPF-extracted supply shocks and the second DPCA factor.

Figure 5: Factors (CF filter) - Cross-Correlation

(A) Demand factor

(B) Supply factor



Note: The left panel shows the cross-correlation of the SPF-extracted demand shocks with the first (demand) DPCA factor. The right panel shows the cross-correlation of SPF-extracted supply shocks with the second (supply) DPCA factor. Bars denote the cross-correlation with the filtered supply/demand shocks. Dots denote the cross-correlation with the unfiltered supply/demand shocks.

Finally, to foster our intuition, we look at impulse response functions implied by the alternative supply and demand shocks. Figure A10 show the impulse response functions of GDP and CPI (detrended by Christiano-Fitzgerald filter) to the supply and demand shock. The top row again displays the impulse response functions of GDP and CPI to the demand shock. The bottom row shows the same regarding the supply shock. The impulse response functions have similar dynamics to the ones obtained using DPCA factors. (The shocks were re-scaled to have the same variances as DPCA factors.)

Generally, we believe that it is an important finding that the supply and demand shocks obtained by this alternative identification method are so similar to the extracted DPCA factors. We stress that the SPF data are basically 'real-time' data and do not undergo revisions. Also, the methodology used significantly differs, and we do not explicitly target the business cycle fluctuations in any way. Therefore, we believe that the findings presented are strong evidence supporting the supply and demand interpretation of common factors.

4 Conclusion

In this paper, we focus on the business cycle fluctuations of a large macro-dataset of US data ranging from 1959 (Q4) to 2022 (Q3). In more detail, we use dynamic principal components analysis (DPCA) with various statistical filters to extract common factors. The main contribution is the identification and interpretation of these factors and the subsequent assessment of their impact on selected variables.

Our approach generally represents a simple way of extracting the supply and demand factors. The study empirically shows the different effects of supply and demand factors on GDP and CPI, where we illustrate that the demand factor is the most prominent driver of the economy. The supply factor has a much smaller impact on the real economy but explains about a third of the cyclical movement of inflation. We also illustrate some evidence that the relative importance of these two factors changes over time and we also propagation of these factors in the economy.

In addition, we show that the factors obtained using DPCA generally correspond to the alternative identification of supply and demand shocks using revisions to expectations from Survey of Professional Forecasters (SPF) data regarding inflation and GDP/GNP. This is especially true regarding the demand shocks. Overall, the study allows for a better structural interpretation of business fluctuations. Although, we admit that our approach to identification is very narrative. However, we believe we present strong enough evidence to support this supply and demand interpretation of common factors.

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A Appendix

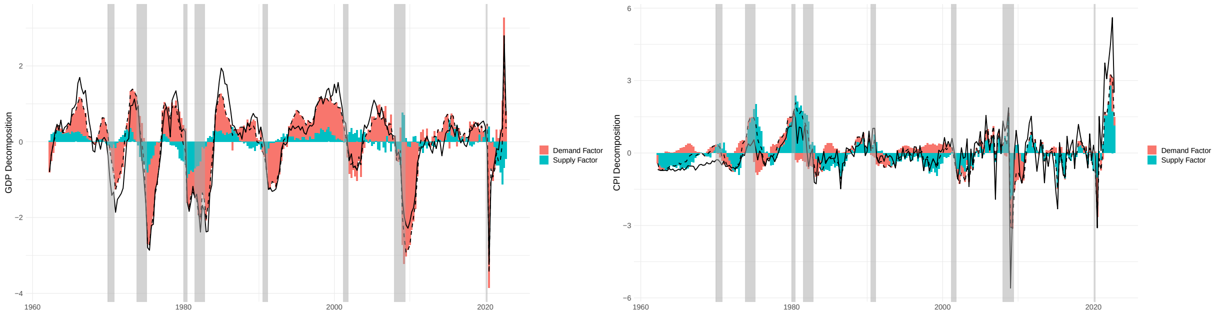
Table A1: Summary Statistics - Categories

CF filter					
Category	**COR_1**	**COR_2**	**RSQ_1**	**RSQ_2**	**RSQ_total**
Earning_Productivity	0.46	0.19	0.24	0.27	0.51
Employment	0.33	-0.25	0.21	0.16	0.38
Exchange_rate	-0.24	0.04	0.43	0.06	0.49
Household_balance_sheet	-0.34	-0.21	0.23	0.14	0.37
Housing	0.38	0.05	0.37	0.13	0.50
Industrial_production	0.13	0.03	0.16	0.08	0.24
Inventories	0.62	0.37	0.44	0.19	0.61
IR	0.21	0.01	0.20	0.06	0.27
Money	0.29	0.03	0.26	0.05	0.30
NIPA	0.33	-0.01	0.34	0.10	0.43
Non-Household_balance_sheet	0.52	-0.19	0.36	0.10	0.48
Other	0.68	0.20	0.48	0.04	0.51
Prices	0.29	0.09	0.43	0.11	0.52
Stock_market	0.54	-0.07	0.35	0.02	0.37
HP filter					
Category	**COR_1**	**COR_2**	**RSQ_1**	**RSQ_2**	**RSQ_total**
Earning_Productivity	0.46	0.19	0.24	0.27	0.51
Employment	0.33	-0.25	0.21	0.16	0.38
Exchange_rate	-0.25	0.04	0.43	0.06	0.49
Household_balance_sheet	-0.33	-0.21	0.23	0.14	0.37
Housing	0.39	0.04	0.35	0.13	0.47
Industrial_production	0.12	0.02	0.16	0.08	0.23
Inventories	0.59	0.36	0.42	0.19	0.59
IR	0.22	0.00	0.21	0.06	0.28
Money	0.23	-0.03	0.23	0.05	0.28
NIPA	0.33	-0.01	0.33	0.10	0.43
Non-Household_balance_sheet	0.52	-0.19	0.36	0.10	0.48
Other	0.68	0.19	0.48	0.04	0.51
Prices	0.28	0.09	0.42	0.11	0.52
Stock_market	0.54	-0.07	0.36	0.02	0.37
Hamilton filter					
Category	**COR_1**	**COR_2**	**RSQ_1**	**RSQ_2**	**RSQ_total**
Earning_Productivity	0.42	0.03	0.21	0.16	0.38
Employment	0.24	-0.32	0.13	0.22	0.35
Exchange_rate	-0.30	-0.06	0.50	0.03	0.53
Household_balance_sheet	-0.31	-0.26	0.24	0.20	0.44
Housing	0.36	-0.08	0.33	0.10	0.43
Industrial_production	0.13	0.08	0.14	0.10	0.24
Inventories	0.63	0.18	0.43	0.06	0.49
IR	0.27	0.06	0.19	0.09	0.28
Money	0.33	0.00	0.26	0.05	0.31
NIPA	0.28	-0.08	0.25	0.11	0.36
Non-Household_balance_sheet	0.37	-0.29	0.25	0.17	0.42
Other	0.73	0.04	0.55	0.00	0.55
Prices	0.25	0.05	0.39	0.10	0.49
Stock_market	0.55	-0.03	0.37	0.03	0.39

Note: The first and second columns show the average correlation coefficient in the given category regarding the first or second factor. The next two columns display the average R-squared with the first or second factor. The last column show the average unadjusted R-squared obtained by regressing the individual series on both factors.

Figure A1: Decomposition (Hamilton filter)

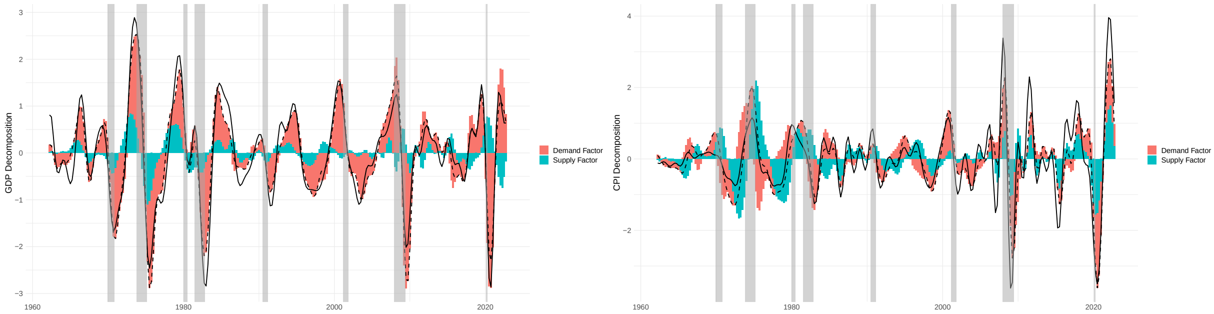
(A) GDP (B) CPI



Note: The dotted line represents a fitted value of GDPC1/CPIAUCSL. Solid line is detrented GDPC1/CPIAUCSL. Shaded regions denote NBER-dated recessions.

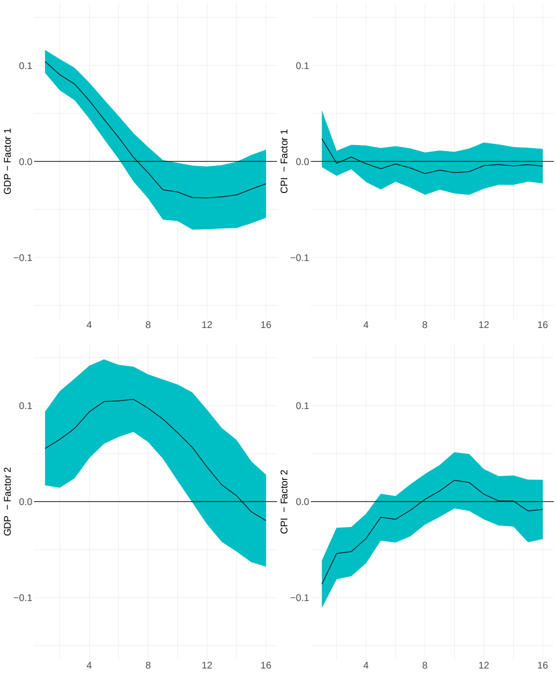
Figure A2: Decomposition (HP filter)

(A) GDP (B) CPI



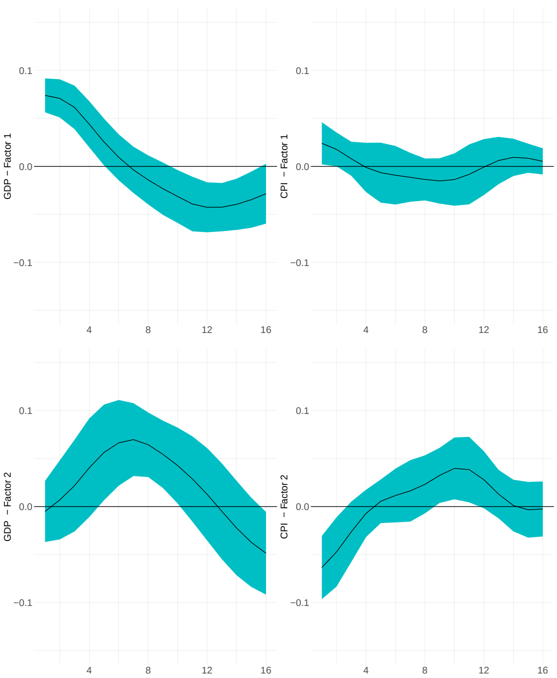
Note: The dotted line represents a fitted value of GDPC1/CPIAUCSL. Solid line is detrented GDPC1/CPIAUCSL. Shaded regions denote NBER-dated recessions.

Figure A3: Impulse response functions (Hamilton filter)



Note: The green bands denote Newey-West robust standard errors (95% confidence intervals). The top row shows the impulse response functions of GDP and CPI to the first (demand) factor. The bottom row shows the impulse response functions of GDP and CPI to the second (supply) factor.

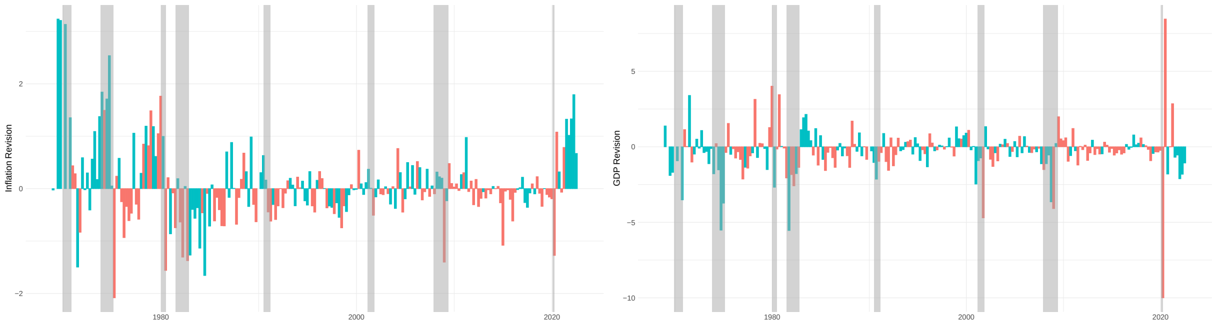
Figure A4: Impulse response functions (HP filter)



Note: The green bands denote Newey-West robust standard errors (95% confidence intervals). The top row shows the impulse response functions of GDP and CPI to the first (demand) factor. The bottom row shows the impulse response functions of GDP and CPI to the second (supply) factor.

Figure A5: Revisions To Expectations

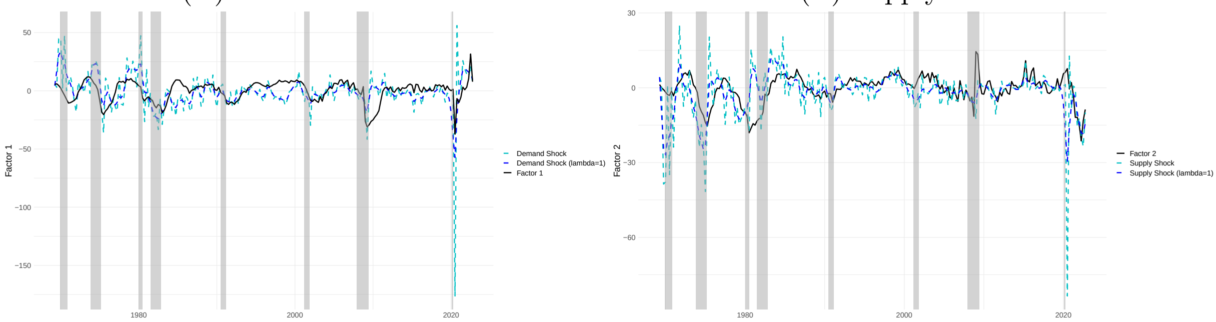
(A) Expectation revisions: GDP/GNP Deflator (B) Expectation Revisions: GDP/GNP



Note: The left panel shows GDP/GNP Deflator revisions. The right panel shows GDP/GNP revisions. Positive co-movement in both series is colored red, e.g., demand shock. Negative co-movement in both series is colored blue, e.g., supply shock. GDP revision was winsorized (1 observation) for illustration purposes. Shaded regions denote NBER-dated recessions.

Figure A6: Factors (Hamilton filter) - Robustness check

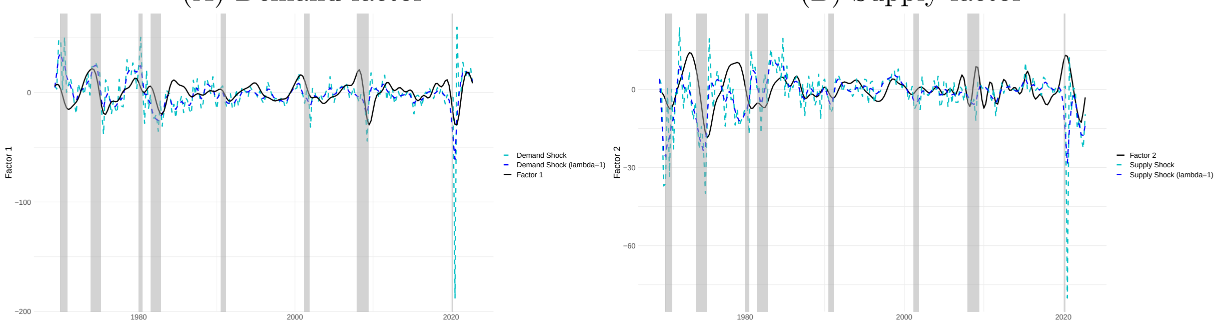
(A) Demand factor (B) Supply factor



Note: The left panel shows extracted demand shocks and the first DPCA factor. The right panel shows extracted supply shocks and the second DPCA factor. supply and demand shocks were obtained from SPF data using revisions to expectations. SPF shocks were re-scaled to have the same variances as DPCA factors. Shaded regions denote NBER-dated recessions.

Figure A7: Factors (HP filter) - Robustness check

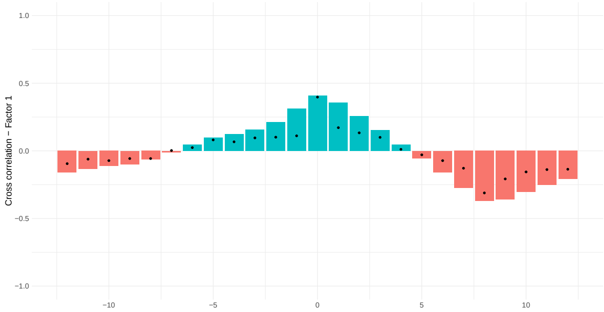
(A) Demand factor (B) Supply factor



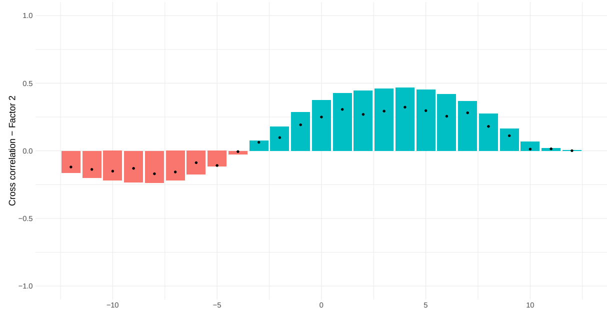
Note: The left panel shows extracted demand shocks and the first DPCA factor. The right panel shows extracted supply shocks and the second DPCA factor. supply and demand shocks were obtained from SPF data using revisions to expectations. SPF shocks were re-scaled to have the same variances as DPCA factors. Shaded regions denote NBER-dated recessions.

Figure A8: Factors (Hamilton filter) - Cross-Correlation

(A) Demand factor



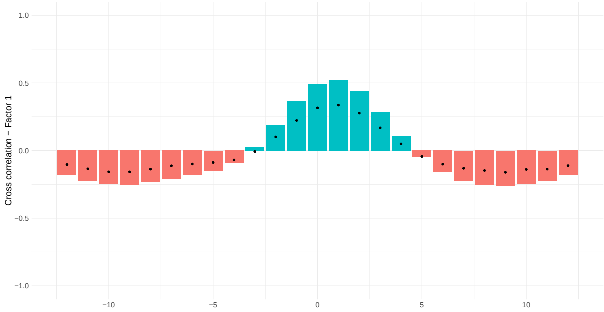
(B) Supply factor



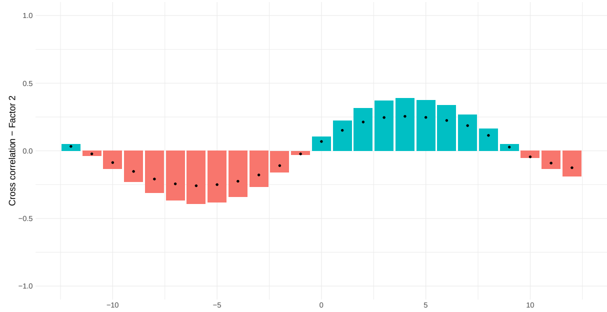
Note: The left panel shows the cross-correlation of the SPF-extracted demand shocks with the first (demand) DPCA factor. The right panel shows the cross-correlation of SPF-extracted supply shocks with the second (supply) DPCA factor. Bars denote the cross-correlation with the filtered supply/demand shocks. Dots denote the cross-correlation with the unfiltered supply/demand shocks.

Figure A9: Factors (HP filter) - Cross-Correlation

(A) Demand factor

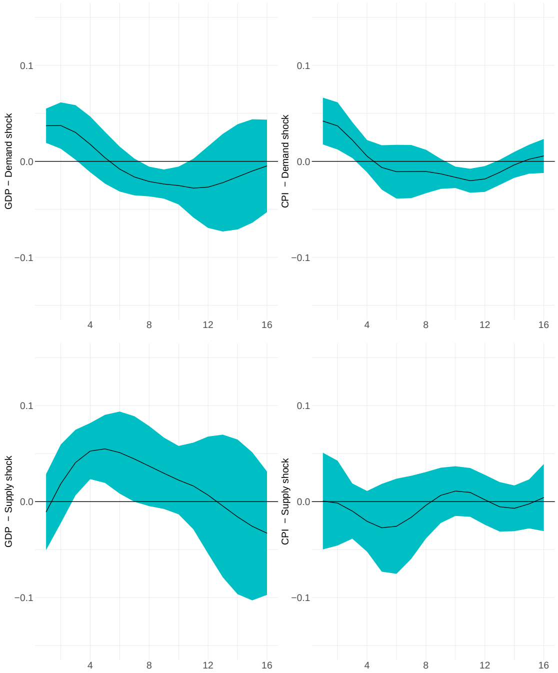


(B) Supply factor



Note: The left panel shows the cross-correlation of the SPF-extracted demand shocks with the first (demand) DPCA factor. The right panel shows the cross-correlation of SPF-extracted supply shocks with the second (supply) DPCA factor. Bars denote the cross-correlation with the filtered supply/demand shocks. Dots denote the cross-correlation with the unfiltered supply/demand shocks.

Figure A10: Impulse response functions (Revisions To Expectations)



Note: The green bands denote Newey-West robust standard errors (95% confidence intervals). The top row shows the impulse response functions of GDP and CPI to the demand shock. The bottom row shows the impulse response functions of GDP and CPI to the supply shock. The supply and demand shocks were obtained from SPF data using revisions to expectations. SPF shocks were re-scaled to have the same variances as DPCA factors. The GDP and CPI were detrended by the Christiano-Fitzgerald filter.